

KAUNAS INTL, Lithuania

WMO: 266295

Lat: 54.964N

Lon: 24.085E

Elev: 256

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.0	(c) 4.7	(d) -7.3	(e) 3.8	(f) -0.1	(g) -2.0	(h) 5.0	(i) 5.2	(j) 26.1	(k) 38.6	(l) 22.6	(m) 37.5	(n) 4.3	(o) 80	(p) 0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	84.4	68.0	80.8	66.7	77.4	64.7	70.6	79.9	68.8	77.5	66.9	74.8	8.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.9	103.4	75.0	65.9	96.5	72.6	64.1	90.4	71.1	34.8	80.2	33.0	77.4	31.5	76.0	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.2	(b) 17.0	(c) 15.5	(e) -6.4	(f) 89.4	(g) 6.7	(h) 4.0	(i) -11.2	(j) 92.3	(k) -15.1	(l) 94.7	(m) -18.9	(n) 96.9	(o) -23.8	(p) 99.9		
(4) 19.2	(5) 17.0	(6) 15.5	(7) -6.9	(8) 73.8	(9) 6.5	(10) 1.6	(11) -11.6	(12) 75.0	(13) -15.4	(14) 76.0	(15) -19.0	(16) 76.9	(17) -23.8	(18) 78.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.0	24.8	26.8	34.7	45.9	55.5	61.3	65.7	64.4	56.3	45.7	38.4	31.8		
	DBStd	15.92	11.24	10.29	8.19	7.33	7.32	6.06	5.28	5.48	5.61	6.69	6.67	8.30		
	HDD50	3189	782	650	477	162	26	1	0	0	13	166	348	564		
	HDD65	7109	1247	1070	940	572	309	144	56	80	266	598	798	1029		
	CDD50	1741	0	0	1	41	195	339	485	446	200	34	0	0		
	CDD65	186	0	0	0	0	13	32	76	61	4	0	0	0		
	CDH74	1678	0	0	0	13	146	344	632	508	35	0	0	0		
	CDH80	400	0	0	0	1	21	74	162	137	5	0	0	0		
(14) Wind	WSAvg	7.6	8.7	8.2	8.3	7.6	7.2	6.9	6.5	6.4	6.7	7.5	8.4	8.7		
(15) Precipitation	PrecAvg	27.1	2.0	1.6	1.5	1.5	2.4	2.8	3.6	3.0	2.2	2.3	2.0	2.0		
	PrecMax	34.5	3.9	3.1	3.5	3.1	4.0	6.7	8.1	6.5	5.0	6.2	4.0	3.0		
	PrecMin	19.3	0.6	0.5	0.5	0.3	1.1	0.5	0.3	0.2	0.7	0.4	0.4	0.8		
	PrecStd	3.3	0.8	0.6	0.7	0.7	0.8	1.1	1.9	1.6	1.1	1.2	0.8	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.0	46.3	59.4	75.4	82.6	86.4	89.3	89.7	78.8	65.9	53.8	48.6		
		MCWB	46.2	42.6	46.9	57.8	65.1	67.8	69.4	70.8	65.3	57.4	49.6	46.7		
	2%	DB	42.9	42.7	53.8	69.6	78.4	82.0	84.4	83.8	73.2	61.1	50.4	46.4		
		MCWB	41.2	39.3	43.9	53.6	62.7	66.2	68.2	69.0	62.3	55.9	48.2	44.4		
	5%	DB	40.9	39.5	49.7	64.5	73.7	77.4	80.7	79.1	69.6	57.6	48.5	43.2		
		MCWB	39.2	37.3	43.2	50.2	59.7	63.3	67.6	66.1	61.0	53.3	47.0	41.7		
	10%	DB	37.7	37.6	45.0	60.4	70.0	73.8	77.1	75.5	66.2	55.4	46.7	41.3		
		MCWB	36.6	35.8	40.8	48.6	58.3	61.6	65.8	64.9	58.5	51.9	45.4	39.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.7	42.3	49.2	59.6	67.7	70.5	74.2	74.3	67.1	59.5	51.2	47.2		
		MCDB	47.6	45.2	56.5	73.7	79.1	81.3	84.4	84.4	76.2	64.2	52.5	48.1		
	2%	WB	41.2	39.5	46.1	55.0	64.4	67.8	71.3	70.2	63.9	56.4	49.0	44.4		
		MCDB	42.3	41.3	51.2	65.8	74.5	78.2	79.8	79.8	71.0	60.3	50.0	45.4		
	5%	WB	38.7	37.4	43.5	52.0	62.2	65.5	69.3	68.1	61.8	54.4	47.0	41.8		
		MCDB	39.9	39.0	47.9	61.6	71.7	74.9	78.0	76.0	67.5	57.8	48.1	43.1		
	10%	WB	36.5	35.8	40.8	49.7	59.6	63.4	67.4	65.9	59.8	52.6	45.3	39.6		
		MCDB	37.5	37.2	44.1	58.0	67.7	71.9	74.1	74.0	65.1	55.4	46.3	40.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	9.1	12.8	18.8	19.7	18.8	17.5	17.7	15.9	11.9	6.8	6.3		
		MCDBR	7.7	9.0	19.3	27.1	24.8	24.3	23.1	24.0	21.1	14.8	8.7	7.0		
		MCWBR	7.0	7.2	12.0	14.4	12.5	11.8	10.4	11.2	12.2	10.0	7.3	6.7		
	5% WB	MCDBR	6.7	7.9	15.9	23.0	22.3	20.6	19.9	20.5	18.3	13.3	7.9	6.9		
		MCWBR	6.5	6.8	11.0	13.7	12.5	11.1	10.1	10.8	11.6	9.5	7.2	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.331	0.367	0.409	0.396	0.388	0.424	0.419	0.382	0.357	0.338	0.300		
	taud		2.196	2.180	2.211	2.193	2.292	2.343	2.262	2.273	2.349	2.396	2.279	2.231		
	Ebn at Noon		196	231	252	257	269	272	259	252	246	224	180	168		
	Edh at Noon		19	28	34	40	38	37	39	37	30	23	18	15		
(44) All-Sky Solar Radiation	RadAvg		181	420	826	1289	1684	1754	1637	1373	947	464	167	113		
	RadStd		36	80	130	164	163	148	159	132	103	88	22	18		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (17 neighbors)		+0.74	N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A

Nomenclature: See separate page